Scholarship Intelligence Platform — Architectural Technical Note

Author: AI Engineering Team | **System:** Scholarship OS v1.0 | **Environment:** Python 3.12 / SQLite / Streamlit

1. System Architecture & End-to-End Pipeline

The system implements an autonomous, polite intelligence crawler engineered to discover, ingest, verify, and monitor Indian scholarships across Government portals (.gov.in/.nic.in), Universities (.ac.in/.edu.in), and Corporate CSR Foundations without human intervention.

Discovery Engine	HTTP Fetcher	Extraction Pipeline	Verification Engine	Change Drift Engine
DuckDuckGo API	Robots.txt parser	Regex + Dateparser	100-Point Scoring	SHA-1 Fingerprint
Keyword Scoring	Rate-limiting (2.5s)	Verbatim Quotes	Deterministic Gates	Diff Tracking
Allowlist Filter	SHA-256 Snapshots	Optional Ollama LLM	Zero Hallucination	Lifecycle Status

2. Anti-Hallucination & Verbatim Grounding Methodology

A core failure mode in AI crawlers is LLM hallucination of financial amounts and deadlines. This platform implements a strict two-pass invariant rule:

- **Deterministic Extraction First:** Regex tokenizers for currency (INR/Rs/Lakhs), deadlines, and age brackets parse text deterministically.
- **Verbatim Grounding Check:** For every extracted field, the system captures an exact context sentence quote. A quote normalization function validates that norm(quote) in norm(raw_snapshot). If false, the field is rejected and assigned **NULL**.
- **Aggregators as Leads Only:** Platforms like Buddy4Study are restricted strictly to discovery; values are never extracted from aggregator pages directly.

3. 100-Point Deterministic Confidence Scoring Rubric

#	Verification Metric	Max Points	Evaluation Logic / Deterministic Rule
1	Official Source Domain	20 pts	Government (.gov/.nic) = 20, Allowlisted Foundation/CSR = 16, Unknown = 0
2	Active Presence on Page	15 pts	HTTP 200 + Fuzzy token match (rapidfuzz >= 90) on headings
3	Application Portal Live	10 pts	Direct application link on official domain & HTTP HEAD reachable
4	Eligibility Grounded	15 pts	(Count of grounded eligibility fields / Extracted fields) * 15
5	Deadline Verifiable	10 pts	Closing date quote grounded verbatim = 10; Rolling cycle = 8; Absent = 0
6	Freshness / Age	5 pts	Snapshot age <= 7 days = 5; <= 30 days = 3; Older = 0
7	No Conflicting Sources	10 pts	No contradictions between multiple circulars/notifications = 10; Else 0
8	Extraction Consistency	10 pts	Deterministic regex pass matches LLM extraction = 10; Partial = Fraction
9	Traceability Coverage	5 pts	Percentage of core fields with linked cryptographic evidence quotes
	Total Maximum Trust Score	**100 pts**	**Status = VERIFIED requires >= 95.0 pts AND all hard gates cleared**

4. Fingerprinting, Change Tracking & Lifecycle State Machine

Each entity is assigned an immutable cryptographic fingerprint: SHA1(norm(name) + norm(provider) + domain). On re-crawling, the system detects field modifications (tracked: closing date, amount, income limit) and records an immutable diff row in change_log.

- **ACTIVE:** Official page verified, closing date in future (> 15 days) or rolling cycle.
- **EXPIRING_SOON:** Closing date within 15 days of current evaluation date.
- **EXPIRED:** Closing date < current date, or notification explicitly marked closed.
- **NO_LONGER_VERIFIABLE:** URL returns 404/500 across two consecutive scheduled cycles.